Submodule 33.02: Handling Missing Values (Median Imputation on Income)

Topic Overview & Mathematical Rationale

In unsupervised distance-based algorithms (such as K-Means, Agglomerative Hierarchical Clustering, or PCA), distance metrics such as Euclidean distance:

$$d(u, v) = \sqrt{\sum_{i=1}^{p} (u_i - v_i)^2}$$

cannot be evaluated if even a single attribute value contains `NaN`. Consequently, missing data must be rigorously imputed prior to running vector space transformations.

In our audit, `Income` contained exactly 24 missing values (~1.07% of the dataset).

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1. Mean vs Median Imputation in Financial Data

Financial distributions such as customer income exhibit strong positive right-skewness due to high earners.

- **Arithmetic Mean (μ)**: Highly sensitive to extreme values. A single multi-millionaire drastically inflates μ.

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- **Median (M)**: Robust order statistic representing the true 50th percentile (central tendency), completely unaffected by skewness:

$$M = \begin{cases} X_{\left(\frac{n+1}{2}\right)} & \text{if } n \text{ is odd} \\ \frac{X_{\left(\frac{n}{2}\right)} + X_{\left(\frac{n}{2}+1\right)}}{2} & \text{if } n \text{ is even} \end{cases}$$

Because household income is naturally skewed and contains extreme outliers, **median imputation** is the statistically sound strategy.

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2. Python Code Implementation

Reference: smartcart.ipynb (Cells 05 - 08).

```
# Impute missing Income values using the distribution median
df["Income"] = df["Income"].fillna(df["Income"].median())

# Verify zero remaining null values
print("Remaining null values in Income:", df["Income"].isnull().sum()) # Output: 0
print("Total dataset null values:", df.isnull().sum().sum())          # Output: 0
```

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3. Verification & Impact

1. Missing count dropped from 24 to 0.
2. Preserved the full sample size ($N = 2,240$) without discarding customer records.
3. Avoided introducing artificial variance into the distance matrix.